

Preferential Transport of Nitrate through Soil Columns Containing Root Channels

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ABSTRACT

Preferential flow has been increasingly recognized as an important mechanism for water and solute transport. The objectives of this study were to characterize transport of NO_3 through root channels in uniformly packed soil columns and to compare three commonly used solute transport models in describing NO_3 transport through the soil-root channel columns. Root channels were created by growing corn (*Zea mays* L.) or alfalfa (*Medicago sativa* L.) in 0.6-m-long soil columns. Transport of NO_3 was then systematically studied with each of these macropore systems at a range of fluxes (from $0.042K_{sm}$ to $0.47K_{sm}$, where K_{sm} is the saturated hydraulic conductivity of the macroporous system). A nonlinear least-squares program was used to fit the convection-dispersion equation (CDE), the physical nonequilibrium model (MIM) and a stochastic model (SM) to the experimental data. The results show that significant preferential movement of NO_3 occurred in the decayed root channels at fluxes as low as $0.042K_{sm}$. The MIM model provided a better description of the breakthrough curves than the CDE and the SM, while none of them were adequate to predict the preferential movement of the solute.

CONCERN OVER groundwater pollution from nonpoint sources, primarily from pesticide and NO_3 leaching in agricultural fields, has greatly increased in recent years. Although the applications of agricultural chemicals have greatly increased the productivity of modern agriculture, deleterious effects of these chemicals on groundwater quality has been reported. Many agricultural chemicals have been detected in the groundwater in many areas of the USA (Holden, 1986; Ritter, 1990; Kanwar, 1991).

Many processes are known to contribute to groundwater contamination (Jaynes et al., 1991). The application of agricultural chemicals to soils may contribute to groundwater pollution via several physical and chemical processes. Extensive groundwater monitoring and experimental data, however, show that preferential flow is likely to be one of the principal mechanisms responsible for accelerated movement of these chemicals in many field soils (Jury et al., 1986; Glass et al., 1988; Ghodrati and Jury, 1990, 1992; Ritter, 1990; Steenhuis et al., 1990; Ellsworth et al., 1991; Kanwar, 1991).

Preferential flow refers to any transport process through which water and solutes gain enhanced movement (Luxmoore, 1991). Preferential flow can be initiated under a number of different conditions, such as flow through macropores (Watson and Luxmoore, 1986), fingering of solute plumes as a result of fluid instability formations (Glass et al., 1988), rapid transport through zones of high hydraulic conductivities (Kung, 1990), or attachment of contaminants to some mobile constituent

of the soil solution (Vinten et al., 1983; McDowell-Boyer et al., 1985). In many agricultural soils, however, macropore flow has been identified as an important mode of preferential flow (Beven and Germann, 1982). Macropore flow could be more significant in Delaware soils because no-tillage practice is prevalent in the state and biologically induced macropores are more abundant in no-tillage soils than in conventional tillage soils.

Macropores are usually described as relatively large and more or less continuous voids (Nielsen et al., 1986) through which rapid flow can occur. Although there is no universal classification, macropores can be roughly divided into two categories: nonbiologically induced and biologically induced. The former refers to the pores formed by soil physical forces. They include interaggregate pores, interpedal voids, cracks and fissures resulting from shrinkage, and natural soil pipes formed by erosive action (Beven and Germann, 1982). Biologically induced macropores represent those formed as a result of biological activities such as soil animal boring and plant growth. Included in this category are root channels, worm holes, ant holes, and other fauna burrows. The majority of field soils contain more than one type of macropore, with their abundance, distribution, and continuity being largely dependent on the tillage practices under which the soils are managed (Ehlers, 1975; Edwards and Loft, 1977; Luxmoore, 1991).

Experimental evidence shows that biologically induced macropores play an important role in water and solute transport (Shipitalo and Protz, 1987; Edwards et al., 1990, 1993), especially in soils under conservation-tillage management, under which the biopores are more abundant and less disturbed. Studying infiltration in different tillage plots, Ehlers (1975) measured a sixfold increase in infiltration rate in a no-till plot compared with the infiltration rate of a nearby tilled plot. He also found that the number and volume percentage of earthworm channels in the A_p horizon approximately doubled during 4 yr of no-tillage practice. Shirmohammadi and Skaggs (1985) found that a 2-yr-old fescue (*Festuca arundinacea* Schreb.) root system increased saturated hydraulic conductivity of a fine sandy soil by fivefold. Bicki et al. (1988) performed infiltration tests in Iowa and attributed the high infiltration rate in one of the soils (an Alfisol) to large-diameter root channels. Preferential flow of water and solutes through root channels has been also observed directly by the application of rhodamine dye (Sollins and Radulovich, 1988).

Recent experiments reported by Ellsworth et al. (1991) have further demonstrated the importance of root channels in solute transport. They studied Cl^- , NO_3^- , and B leaching in 12 soil plots, where the surface soil was

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thoroughly disturbed to 8-cm depth in 11 plots and one plot was left undisturbed. The movement of these solutes was successfully predicted with a two-parameter model as a function of depth and time for all disturbed plots but not for the undisturbed one. The much deeper movement of the solutes at the undisturbed plot was ascribed to the presence of root channels.

Solute transport studies with undisturbed soil cores, lysimeters, or in situ techniques have proven to be useful in characterizing macropore flow. Most field soils or undisturbed soils, however, are complex porous media containing many different types of macropores and other variabilities. Many of these macropores may contribute to the observed preferential flow. To better evaluate the specific role of a certain type of macropore in preferential flow, it is essential to isolate that macropore, eliminate other variabilities, and examine that macropore individually. However, studies where preferential flow through a particular type of macropore is considered are few (Gish and Jury, 1983). Systematic characterization of preferential solute transport in biologically induced macropores, such as root channels, is lacking. The experimental data needed to validate existing solute transport models are limited and the development of more refined approaches for modeling solute preferential flow is also hindered due to the lack of experimental data (Butters et al., 1989).

Several theoretical models, including the CDE, the MIM, and SM, have been developed to describe the one-dimensional solute transport in both laboratory and field situations. The CDE assumes steady-state water flow and a constant water content. Convective flow is described as an "apparent flux," which averages numerous local flow velocities. An overall dispersion coefficient, D , is used to characterize the total effects of diffusion and hydrodynamic dispersion. Although the CDE has been shown to provide good description of some solute transport in field soils or undisturbed soil columns (Parker, 1984; Seyfried and Rao, 1987), its success is generally limited to homogeneous soils. In attempts to improve the predictive capabilities of the CDE, van Genuchten and Wierenga (1976) extended the mobile and immobile conceptualization developed by Coats and Smith (1964) to solute transport in soils. The resulting model is known as the physical nonequilibrium or MIM model. This model assumes the presence of two soil-water regions in solute transport: a mobile or dynamic region and an immobile or stagnant region. Convective-dispersive solute transport is limited to the mobile region only. The ratio of the mobile water content over the total water content, β , is employed to describe the fraction of water that participates in the solute transport. Solute transfer between the two regions is assumed to be diffusion controlled. A mass transfer coefficient, α , is used to characterize the rate of solute transfer between the two regions. Solute dispersion in the mobile region is represented by a mobile dispersion coefficient (D_m). This model has been shown to be superior to the CDE in describing the frequently observed early breakthrough and tailing in packed and undisturbed soil columns (van Genuchten and Wierenga, 1976, 1977;

van Genuchten et al., 1977; Rao et al., 1980; Seyfried and Rao, 1987; De Smedt and Wierenga, 1984; De Smedt et al., 1986).

Due to the highly heterogeneous nature of field soils, the actual mechanisms involved in local flow may not be revealed exactly. Stochastic models, which ignore the actual mechanism of water flow and transport and treat the transport process as a "black box," have been developed to respond to this problem (Bresler and Dagan, 1981; Amoozegar-Fard et al., 1982; Jury, 1982). In the models of Bresler and Dagan (1981) and Amoozegar-Fard et al. (1982) the transport region is considered to be composed of numerous parallel porous columns, each having its specific properties and boundary conditions. The dispersion coefficient is assumed to be proportional to pore water velocity (v), both of which are considered to be log-normally distributed. The local dispersivity, λ , is assumed to be deterministic and constant with each flow region. The soil properties within each column are assumed to be uniform and the one-dimensional CDE is assumed valid to describe the solute transport through each column.

The objectives of this study were to: (i) characterize the preferential flow of NO_3 in decayed root channels and (ii) compare three solute transport models, the CDE, the MIM, and the SM reformulated by Parker and van Genuchten (1984) with respect to their ability to describe NO_3 movement in the soil-root channel columns.

MATERIALS AND METHODS

In order to isolate root channels from other types of macropores, column techniques were used to generate root channels, and miscible displacement techniques were employed to investigate preferential movement of NO_3 through the root channels in the soil columns. The general procedure involved packing previously air-dried soil samples into large acrylic plastic columns (0.2-m i.d. and 0.6 m long) and generating root channels by growing plants on these soil columns. After the roots were decayed, the columns were moved to the laboratory for leaching study. The transport behavior of NO_3 in uniformly packed soil columns was first investigated. Then, similar leaching experiments were conducted in soil columns containing root systems at a wide range of water fluxes to simulate unsaturated flow conditions. Specifically, for each column, the saturated hydraulic conductivity, K_s (no root system) or K_{sm} (with root channels), was measured and then leaching experiments were performed at fluxes varying from 0.042 to $0.47K_{sm}$. This allowed us to examine the contribution of macropores to preferential transport of NO_3 at a wide range of water flow rates in the soil columns.

The soil used in this study was collected from a strip of fallow land on the University of Delaware Farm that was mapped as a Matapeake silt loam (fine-loamy, mixed, mesic Typic Hapludult). The Matapeake series is widely distributed in the northern part of Delaware and is one of the most productive agricultural soils in the state. Soil samples taken from the Ap horizon were sieved through a 2-mm screen at a water content of approximately 150 g kg^{-1} and then air dried. Subsamples were analyzed for selected physical and chemical properties, which are summarized in Table 1.

Acrylic plastic tubes were used to make the soil columns. The tubes were first cut longitudinally into two halves, 0.75 m long. Ridges of silicone sealant were laid in 0.05-m increments

Table 1. Selected properties of Matapeake silt loam soil.

Texture, %	
Sand	31
Silt	53
Clay	16
Bulk density, Mg m^{-3}	1.25
Organic matter, g kg^{-1}	34
pH (in 0.01 M CaCl_2 , 1:2 soil/solution)	5.5
Anion-exchange capacity, $\text{cmol}_c \text{ kg}^{-1}$	0

on the interior surface of each section to eliminate side flow. These two halves were then assembled with silicone sealant and the columns were reinforced by placing two bracelets along their lengths. The columns were capped with a perforated stainless steel plate with a mesh lining.

Six columns were packed uniformly to a bulk density of 1.25 Mg m^{-3} , two for control (referred to as CK-1 and CK-2), two for generating alfalfa root channels (A-1 and A-2), and the other two for creating corn root channels (C-1 and C-2). All columns were packed to a depth of 0.6 m, leaving the top 0.15 m for solution application setup.

Four of the packed columns were moved to the greenhouse and wetted with tap water to near field capacity. The wetting process was accomplished with a precise multichannel peristaltic pump (AP-200, Orion Scientific Instruments Co., Hawthorne, NY)¹ at low flux. Columns C-1 and C-2 were sown with five corn seeds, which were later thinned to two seedlings. Columns A-1 and A-2 were each sown with 10 alfalfa seeds. All columns were irrigated daily with a fine spray nozzle to near saturation. For corn columns, N was applied once as KNO_3 at a rate of 50 kg/ha when the plants were 0.15 m tall. After 60 d of growth for corn and 90 d of growth for alfalfa, irrigation was stopped, and the plants were allowed to dry before the tops were removed at the soil surface. The columns were then wetted and left in the greenhouse for 18 mo to allow the roots to decompose.

To ensure uniform solution application and leaching, a multichannel peristaltic pump was employed that applied the solution at 19 equally spaced points on the soil surface. Each column was first wetted from the top at a very low flux to minimize air entrapment. The flux was then increased progressively and the pump was adjusted several times until a constant thin layer of water (2 mm) appeared. This flux was taken as saturated hydraulic conductivity.

For breakthrough studies, each column was leached with 6 mM CaCl_2 solution at the desired flux until steady-state water flow conditions were achieved. A pulse (0.5 cm) of 60 mM $\text{Ca}(\text{NO}_3)_2$ solution was then introduced to the column at the same flux and the leaching regime was resumed. A fraction collector was used to sample the effluent. Average pore volume (V_p) for each column at each run was determined by maintaining a water balance and by weighing the column before and after sampling; the average value was used to normalize the pore volume.

The effluent was analyzed for NO_3 with a Technicon Auto Analyzer (Technicon Industrial Systems, Tarrytown, NY). To prevent chemical transformation of NO_3 by microbial activities, the analysis was done immediately after sampling. The NO_3 mass recovery for all of the displacement experiments was >95%. Breakthrough curves were generated by plotting relative concentration, C/C_0 , vs. number of pore volumes, V/V_0 , where C and C_0 are NO_3 concentrations in the effluent and in the pulse and V and V_0 represent effluent volume and pore volume, respectively.

¹ Mention of trade names is solely to provide specific information and does not constitute an endorsement over similar products not mentioned.

To test the ability of the three solute transport models (CDE, MIM and SM) to describe NO_3 transport in the soil columns containing root channels, a nonlinear least-squares analysis program CXTFIT (Parker and van Genuchten, 1984) was used to fit all three models to the experimental data for all treatments. The best fit (least sum of squares) was obtained by adjusting the relevant parameters in corresponding models. All three models have a retardation factor, R , which characterizes the degree to which a solute applied to a soil is retarded relative to a conservative tracer. The R value for NO_3 was fixed as 1, since the anion-exchange capacity of the Ap horizon of the soil was zero (Table 1). The overall dispersion coefficient (D) was fitted to the CDE. The mobile dispersion coefficient (D_m), the fraction of mobile water content (β), and mass transfer coefficient (α) were fitted to the MIM. Dispersivity (λ) and time-averaged pore water velocity (v) were fitted to the SM.

RESULTS AND DISCUSSION

Saturated Hydraulic Conductivity

The saturated hydraulic conductivity, K_s or K_{sm} , for each column was measured prior to the displacement experiments. When compared with the control columns, an increase in saturated hydraulic conductivity of the corn or alfalfa columns would indicate the alteration of the uniformity of the columns and the development of the root channel macropores. In addition, the values of the saturated hydraulic conductivity were also used as a guide for choosing the range of flow rates that were implemented in our study to evaluate preferential movement of water and NO_3 through the root channel columns. The K_s values for the two control columns, CK-1 and CK-2, were 7.92×10^{-6} and $8.39 \times 10^{-6} \text{ m s}^{-1}$, respectively, while the corresponding K_{sm} values for the two alfalfa columns were 5.89×10^{-5} and $4.78 \times 10^{-5} \text{ m s}^{-1}$, and for the corn columns were 5.44×10^{-5} and $4.92 \times 10^{-5} \text{ m s}^{-1}$. The large K_{sm} values of the root channel columns (6–7 times greater than the K_s of the control columns) indicate that both alfalfa and corn root channels have created relatively continuous macropores that have extended to the bottom of the soil columns. The differences in K_{sm} values between the columns containing the two types of root channels were not significant.

Evaluation of Breakthrough Curves

Figure 1 shows NO_3 BTCs for the two control columns, which were leached at a similar relative flow rate. The BTC from column CK-1, which was generated at a flow rate of $4.44 \times 10^{-6} \text{ m s}^{-1}$, is characterized by a symmetric distribution. Nitrate did not break through until 0.5 pore volume of leaching solution had percolated, and the peak concentration occurred at about one pore volume. The BTC for CK-2 at a flow rate of $4.31 \times 10^{-6} \text{ m s}^{-1}$ exhibited a similar shape. No NO_3 appeared in effluent before 0.56 pore volume. It is apparent that even at these relatively high fluxes, the shapes of these BTCs for the control columns are still relatively symmetric. This indicated that little or no preferential flow took place in these columns.

Contrastingly, the NO_3 BTCs for the root channel columns are rather skewed (Fig. 2 and 3), indicating that the decayed roots had formed well-connected channel

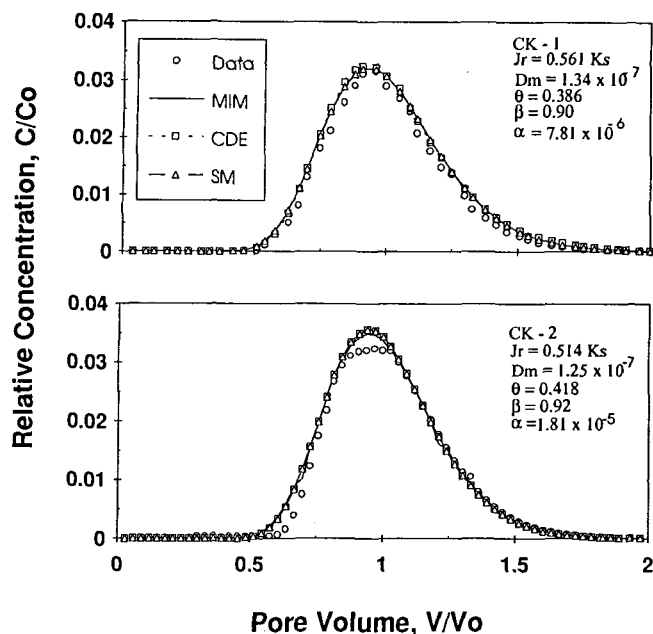


Fig. 1. Observed and fitted breakthrough curves from the two control columns. Data represent the experimental breakthrough curve, J_r is the relative flux, θ is the volumetric water content ($\text{m}^3 \text{m}^{-3}$) of the soil column at the specified flux, and other variables are the parameters fitted from the mobile and immobile model.

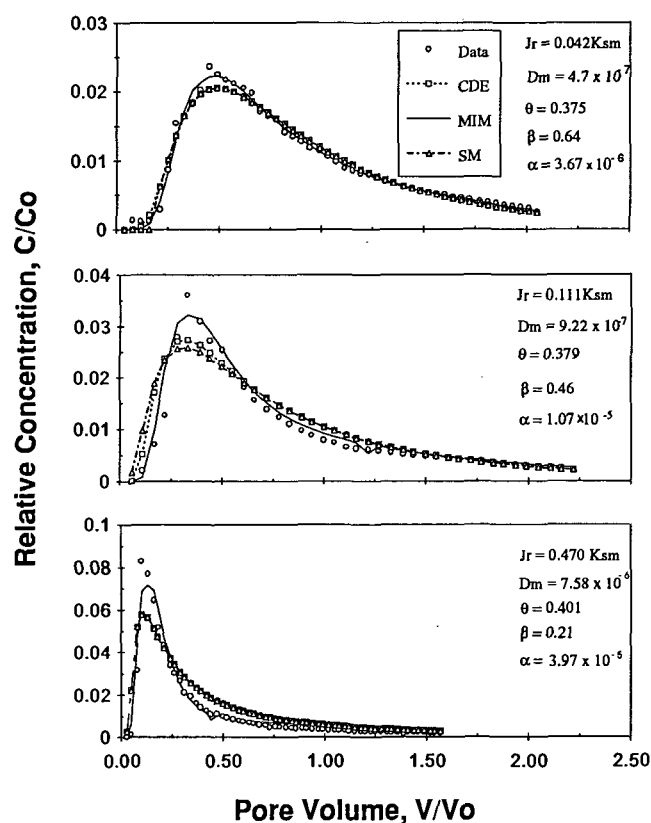


Fig. 2. Observed and fitted breakthrough curves from alfalfa root column A-1 at three flow rates. Data represent the experimental breakthrough curve, J_r is the relative flux, θ is the volumetric water content ($\text{m}^3 \text{m}^{-3}$) of the soil column at the specified flux, and other variables are the parameters fitted from the mobile and immobile model.

systems in the soil columns, allowing the preferential movement of NO_3^- . Most of the BTCs of alfalfa root channel columns (Fig. 2) are characterized by a much earlier breakthrough and a greater extent of dispersion than the control columns. The BTCs became progressively more skewed, with considerable tailing, and shifted to the left as flow rates increased. Three of the five NO_3^- leaching experiments conducted in column A-1 with fluxes ranging from $0.042K_{sm}$ to $0.470K_{sm}$ are shown in Fig. 2. The breakthrough of NO_3^- occurred at 0.2 pore volume even at fluxes as low as $0.042K_{sm}$. Increasing fluxes resulted in even earlier breakthroughs and greater peak concentrations. At the highest flow rate, which was equivalent to $0.470K_{sm}$, NO_3^- appeared in the effluent when only 0.05 pore volume of leaching solution had passed through the column. The peak relative concentration (0.08) occurred at 0.10 pore volume and 80% of the solute had passed through the column when <0.3 pore volume had percolated. At a flux of $0.358K_{sm}$, the BTC (not shown) was almost identical to that of $0.47K_{sm}$, except that the peak relative concentration was about 0.07. This suggests that the root channels formed in this column were almost equally effective in inducing preferential flow at the flux of $0.358K_{sm}$ as with a flux of $0.470K_{sm}$.

Similar to the column A-1, the BTCs generated in the

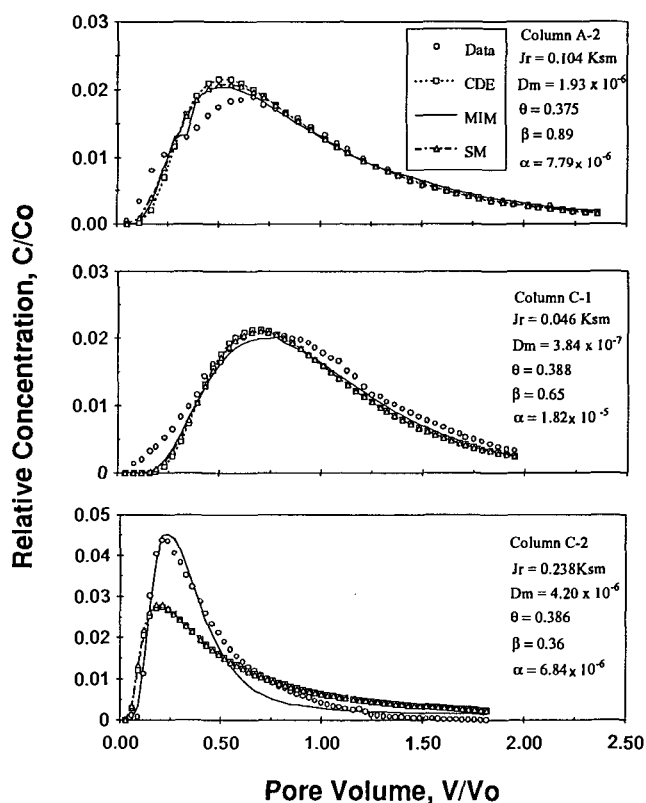


Fig. 3. Observed and fitted breakthrough curves from alfalfa root column A-2 and corn root columns C-1 and C-2. Data represent the experimental breakthrough curve, J_r is the relative flux, θ is the volumetric water content ($\text{m}^3 \text{m}^{-3}$) of the soil column at the specified flux, and other variables are the parameters fitted from the mobile and immobile model.

other alfalfa column and in the two corn columns were all characterized by a rapid shift to the left, and a sharp increase to concentration maxima, followed by a slow decrease in concentration as fluxes increased from $0.046K_{sm}$ to $0.435K_{sm}$ (Table 2, Fig. 3). This indicated that preferential flow was prevalent in both root channel systems at all flow rates used in this study.

Significant preferential movement of solute resulting from root channels has also been observed in decayed wheat (*Triticum aestivum* L.) root channel systems by Gish and Jury (1983). These researchers, however, did not notice any mass transfer between mobile and immobile zones and this was probably due to their use of very low flow rates. In our study, mass transfer became pronounced with increasing flow velocities and this agreed with many other solute transport studies (van Genuchten and Wierenga, 1977; van Genuchten et al., 1977; De Smedit and Wierenga, 1984; De Smedit et al., 1986). It is apparent that, like wheat roots, both alfalfa and corn roots are able to form well-connected macropores that can initiate preferential flow of water and solutes under the leaching conditions used in this study, constituting a major mechanism for the leaching loss of solutes.

Evaluation of the Models and the Parameters from the MIM Model

Figures 1 to 3 show the observed and fitted BTCs from the CDE, the MIM, and the SM models. In the majority of the cases, the MIM fitted the data better than the other two models. However, it is apparent that even for the MIM, the fitting decreased with increasing flux. The fitted parameters obtained from this model are reported in Table 2.

The observed BTCs and those fitted from the three models for the control soil columns are plotted together and illustrated in Fig. 1. The fittings are generally good. All three models seem to fit the data equally well, al-

though the fittings to the BTC of column CK-2 are not as good as column CK-1.

For the root channel columns, the CDE and the SM fitted the data well only at relatively low fluxes. In the presence of root channels, when fluxes increased to $0.1K_{sm}$ or higher, the inadequacy of using these models to describe NO_3 transport became evident (Fig. 2 and 3). This inadequacy in fitting experimental data by the CDE is expected, since it inherently assumes that all the pore water in the column is mobile and participates in solute transport and the average local velocities represent the flow well. These assumptions do not hold when part of the solute transport takes place in macropore systems like root channels. As shown above, preferential flow of NO_3 occurred in the root channels even at flow rates as low as $0.042K_{sm}$ and it became more significant when fluxes increased. Apparently, a portion of the water was immobile and only a fraction of the water participated in the convective flow process. The SM is a stochastic model that treats the soil column as a system composed of numerous local flow domains with random flow processes occurring in each domain. It appeared that in this study the SM, similar to the CDE, was unable to fit the data at high flow rate conditions. Similar results were obtained by White et al. (1984). The MIM described the transport better than both the CDE and the SM, although better at low fluxes than at high fluxes. Khan (1986) tested the validity of the CDE by characterizing Cl^- movement in soil columns of various surface area, length, and flow rates. He also compared the capability of these models to fit the experimental data and found that in nearly all cases the four-parameter MIM model fitted the data much better than the other two. The mobile dispersion coefficient D_m increased greatly with the flux and mobile water velocity (v_m), especially for the root channel columns (Fig. 4). A linear relationship between the D_m and v_m was fitted through the data, ($D_m = 0.0215v_m + 7.14 \times 10^{-7}$) with $r^2 = 0.67$. These large dispersion coefficients indicated that flow through the root channel

Table 2. Experimental conditions and curve-fitting parameters from the mobile and immobile model from the miscible displacement experiments.

Treatment†	Flux, J $m\ s^{-1}$	Relative flux, J_r	Volumetric water content, θ $m^3\ m^{-3}$	Mobile water fraction, β	Mobile water content, θ_m $m^3\ m^{-3}$	Mobile dispersion coefficient D_m $m^2\ s^{-1}$	Mass transfer coefficient, α s^{-1}
CK-1	4.44×10^{-6}	0.561	0.386	0.90	0.347	1.34×10^{-7}	7.81×10^{-6}
CK-2	4.31×10^{-6}	0.514	0.418	0.92	0.385	1.25×10^{-7}	1.81×10^{-5}
A-1-1	2.50×10^{-6}	0.042	0.375	0.64	0.240	4.70×10^{-7}	3.67×10^{-6}
A-1-2	5.31×10^{-6}	0.090	0.377	0.35	0.132	1.57×10^{-6}	1.35×10^{-5}
A-1-3	6.56×10^{-6}	0.111	0.379	0.46	0.174	9.22×10^{-7}	1.07×10^{-5}
A-1-4	2.11×10^{-5}	0.358	0.380	0.24	0.091	2.46×10^{-6}	2.15×10^{-5}
A-1-5	2.77×10^{-5}	0.470	0.401	0.21	0.084	7.58×10^{-6}	3.97×10^{-5}
A-2-1	4.97×10^{-6}	0.104	0.375	0.89	0.334	1.93×10^{-6}	7.79×10^{-6}
A-2-2	1.17×10^{-5}	0.244	0.418	0.33	0.138	2.40×10^{-6}	3.50×10^{-6}
C-1-1	2.53×10^{-6}	0.046	0.388	0.65	0.252	3.84×10^{-7}	1.82×10^{-5}
C-1-2	5.00×10^{-6}	0.092	0.395	0.66	0.261	9.73×10^{-7}	2.50×10^{-5}
C-1-3	1.42×10^{-5}	0.261	0.405	0.33	0.134	4.58×10^{-6}	4.01×10^{-5}
C-2-1	1.17×10^{-5}	0.238	0.386	0.36	0.139	4.20×10^{-6}	6.84×10^{-6}
C-2-2	2.14×10^{-5}	0.435	0.376	0.31	0.117	7.82×10^{-6}	2.32×10^{-5}

† CK-1 and CK-2 are control columns; A-1 and A-2 are replicate soil columns containing alfalfa root channels; and C-1 and C-2 are replicate soil columns containing corn root channels. The last number represents the number of experiments conducted within a column.

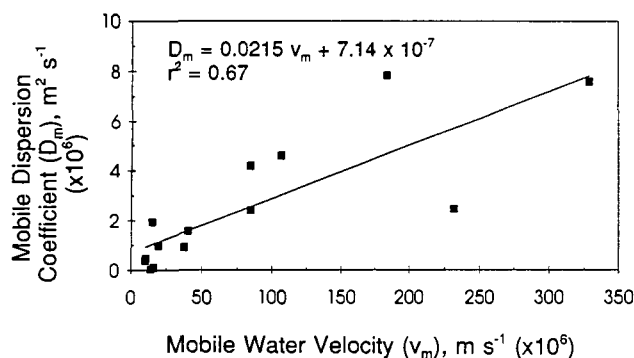


Fig. 4. The relationship between the mobile dispersion coefficient and the mobile water velocity.

columns was highly heterogeneous and this heterogeneity was increased with increasing flow rates. Large dispersion were also reported by Parker (1984) for Br^- movement in worm hole columns and by Seyfried and Rao (1987) for tritiated water in undisturbed soil columns.

The values of the mobile water fraction, β , which characterizes the portion of water that participated in convective flow, were much greater in control columns than in root channel columns (Table 2). However, no unidirectional relation was found between β and the flow rates. The same phenomenon was also observed in other studies (De Smedit and Wierenga, 1984; Seyfried and Rao, 1987) although these researchers obtained a better fitting of these models to their data.

In this study, the mass transfer coefficient, α , refers to the mass of NO_3 diffused from the immobile water region to the mobile water region. Under unsaturated flow conditions, a portion of the water is stagnant. The dissolved NO_3 in this region has to transfer into the mobile region to be transported. A greater α would indicate faster NO_3 transfer between the root channel macropores and the soil matrix. The relationship between α and flux appeared to be complicated in this study. The value of α was found to increase with increasing mobile water velocity. However, the linear regression coefficient is not significantly high. Research by Rao et al. (1980) and Nkedi-Kizza et al. (1984) has shown that for a given ionic species, α is not a constant but a function of flux, aggregate size, β , and the concentration of leaching solution. Theoretically, for a given soil column at a fixed ionic strength, α is proportional to flux q . A linear relationship between α and v_m was found by De Smedit and Wierenga (1984) for Cl^- transport in a uniformly packed glass bead column. As stated above, the fitting of the MIM model to the BTCs at higher flow rates was only fair. Those α values calculated from the MIM for those displacement experiments conducted at high fluxes were simply an approximation of the real values. In addition, it is plausible that the mass transfer between mobile and immobile zones in a complex system such as a macroporous soil may require a more complicated form than a simple first-order relation. More carefully designed displacement experiments such as flow through uniform macropore(s) are needed to explore the mass transfer processes in macropore systems.

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